1.3.1 Voltage

Let us see an example to understand about voltage. In our home, water is filled in the overhead tank using electrical motor pump. Due to atmospheric air pressure and gravitational force, tank water is at the higher pressure. The pressure of the water in the tank is assumed as the supply voltage as shown in Figure 1.4.

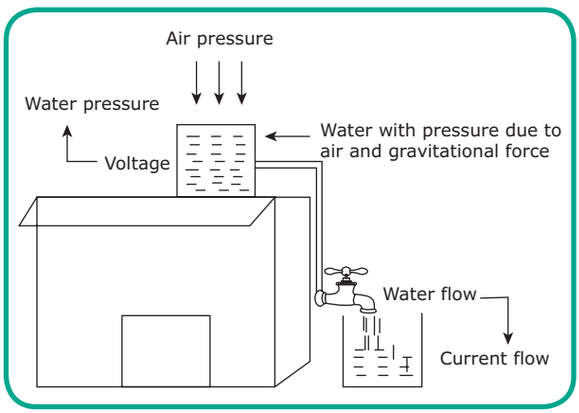


Figure 1.4 Comparison of voltage with pressure of water

The amount of electrical pressure required to pass the current in a conductor is known as voltage. It is represented by the letter V or E and measured by the unit Volt. Highest unit value is kilo-Volt (kV). Lower value units are milli-Volt (mV) and micro-Volt (μV).

The quantity of one volt is defined as the voltage needed to send one ampere of current through one ohm resistance in a circuit.

As we designed weighing stone to measure rice, wheat etc., we defined one volt in the above paragraph.

1.3.2 Electric Current

Let us try to understand current through an example. If we open a tap in our home, water flows quickly because it gets a path to flow from tank where water is under pressure.

Likewise, if we connect an electrical load through proper conducting wires, Electrons will flow from higher density point to lower density point as shown in Figure 1.5. This is called Electric Current.

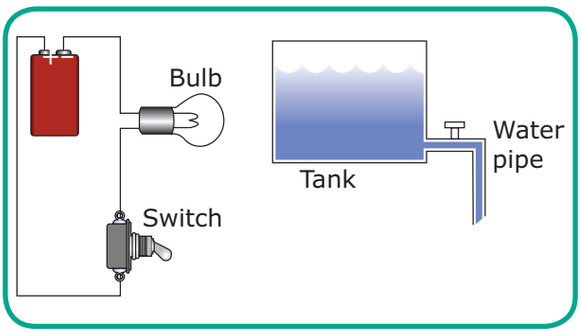


Figure 1.5 Comparison of electric current with flow of water

The flow of current can be measured as 'the number of electrons crossing the particular cross section of a conductor in a second". It is represented by the letter 'I'. Its unit is Ampere. It is the highest value. Lower values are milli-Ampere (mA) and micro-Ampere (μA).

1.3.3 Resistance

The property of a material which opposes the flow of current is known as resistance. An electrical component which has the resistance property is said to be resistor. It is denoted by the letter R and measured by the unit ohm. (The units used to measure higher value resistance are kilo ohm (kΩ) and mega ohm (MΩ).

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